

JULIUS MATHEKA CHARLES

Software Engineer · Data Engineer · AI & ML Developer

✉ cjmatheka@gmail.com

github.com/musee77

linkedin.com/in/julius-charles24b91b89

PROFESSIONAL SUMMARY

Results-driven Software & Data Engineer with hands-on experience across the full data lifecycle — from pipeline design and SQL analytics to Python backend development and BI reporting. Skilled at turning complex data into actionable business insights and building reliable, scalable systems. Experienced working in both internship and junior professional roles, with a strong foundation in Agile delivery, cloud-native tools, and AI/ML integration.

TECHNICAL SKILLS

Data & SQL	PostgreSQL, MySQL, SQL (advanced), ETL/ELT Pipelines, Apache Airflow, Apache Kafka, Data Warehousing, pgvector
BI & Analytics	Power BI, Tableau, Data Modeling, Reporting Dashboards, Business Intelligence, DAX
Python	Django, Django REST Framework, FastAPI, Pandas, NumPy, Celery, pytest, Shell Scripting
AI / ML	LangChain, OpenAI (GPT-4o), TensorFlow, RAG Systems, Vector Embeddings, Semantic Search, Claude API
Frontend	React, Next.js, TypeScript, JavaScript, Tailwind CSS, HTMX, HTML/CSS
DevOps	Docker, GitHub Actions / CI-CD, Linux, Git/GitHub, Agile/Scrum, Redis

PROFESSIONAL EXPERIENCE

Junior Python Data Engineer

Sep 2025 – Present

Maesk Group, Kenya

On-site

- Designed and implemented end-to-end data pipelines in Python and SQL — covering raw ingestion, staging, transformation, and analytics layers across PostgreSQL databases.
- Built and scheduled Apache Airflow DAGs to automate recurring ETL jobs, replacing manual processes and improving data freshness for downstream consumers.
- Engineered a semantic job-matching pipeline using Python, LangChain, and pgvector — parsing CVs and vectorising content for similarity search across 2,400+ live listings.
- Implemented incremental loading patterns (upsert / ON CONFLICT DO UPDATE) and run logging to support efficient, restartable pipeline execution at scale.
- Delivered a RAG-powered knowledge base using PostgreSQL/pgvector and LangChain, achieving sub-3-second query latency under load in production testing.

Junior Python Developer

Aug 2024 – Jun 2025

Paystack

On-Site

- Built scalable Python backend services and data ingestion scripts for small business clients — automating data collection from APIs and CSV sources into PostgreSQL.
- Developed Django-based web applications with user authentication, role-based access, and data export functionality tailored to client reporting needs.
- Integrated LangChain and OpenAI APIs to add AI-powered features including document summarisation, semantic search, and automated content generation.
- Containerised applications with Docker and set up GitHub Actions CI/CD pipelines, enabling reliable and repeatable deployments across client environments.
- Maintained 80%+ test coverage using pytest; resolved bugs and performance bottlenecks identified through production monitoring and client feedback.

Data Analyst Intern (SQL)

Jan 2023 – March 2024

Internship — SQL & Data Analysis

- Wrote complex SQL queries across PostgreSQL databases to extract, clean, and analyse business datasets for weekly management reporting.
- Built and maintained ETL pipelines that pulled data from multiple source systems into a central data warehouse, reducing manual data preparation time by 35%.
- Created data quality checks using SQL (NULL validation, duplicate detection, row count reconciliation) to ensure pipeline integrity across all reporting tables.
- Collaborated with senior analysts to design dashboard-ready summary views and aggregation tables consumed by Power BI reports.
- Documented SQL query logic and pipeline steps, improving team knowledge sharing and onboarding efficiency.

KEY PROJECTS

AI Agent Platform — Agents 47

2024 – Present

github.com/musee77/agents47 | agents47.online

Remote

- Built scalable AI agent platform with RAG knowledge bases (PostgreSQL/pgvector), async pipelines (Celery/Redis), and real-time chat at <3s latency.
- Django 5.x/DRF, PostgreSQL/pgvector, LangChain, OpenAI/Claude, Docker, GitHub Actions: Stack

AI Website Conversion Optimizer — LezMarket

2024

lezmarket.io

Remote

- SaaS tool that scores homepages against persuasion principles, generates AI recommendations, and stores structured analysis history per user.
- Next.js, Supabase (PostgreSQL), OpenAI GPT-4o, TypeScript, Tailwind CSS: Stack

AI Job Matching Platform — Synovae

2024

synovae.io

Remote

- Semantic CV-to-job matching engine across 2,400+ live roles using LLM embeddings, with automated application tracking in Supabase.
- Next.js, Supabase (PostgreSQL), OpenAI/LLMs, TypeScript, Tailwind CSS: Stack

EDUCATION

Professional Certification — Software Engineering

Jan 2024 – Feb 2025

ALX Africa - Full-stack, algorithms, Django/React, Java/Spring Boot

Professional Certification — Data Science

Jan 2023 – Nov 2023

Moringa School - ML, ETL pipelines, Python, SQL, TensorFlow, Tableau

B.Sc. Computer Science

2012 – 2016

Masinde Muliro University of Science & Technology

CERTIFICATIONS

IBM — ETL and Data Pipelines with Shell, Airflow & Kafka

2024

IBM — Data Warehousing and BI Analytics

2024

IBM — Linux Commands and Shell Scripting

2024

REFERENCES

Patrick Mwachiro

- Position: Procurement Officer
- Organisation: Windle.org
- Email: pmwachiro@windle.org

- Relationship: Mentor